

Program : Diploma in Polymer Technology	
Course Code : 4094	Course Title: Plastic Technology
Semester : 4	Credits: No Credit
Course Category: Program Core	
Periods per week: 3 (L:3, T:0, P:0)	Periods per semester: 45

Course Objectives:

- To understand the general characteristics and classification of plastics
- To know the preparation ,properties and application of commodity plastics
- To know the preparation, properties and application of engineering plastic.
- To know the preparation ,properties and application of thermo setting plastics
- To understand compounding and compounding equipment's used in plastics

Course Prerequisites:

Topic	Course code	Course name	Semester
Classification of polymers, chemistry of polymerisation and polymerisation technique		Fundamentals of Polymer Science	2
Polymer reactions, stereo regular polymers		Polymer science	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (hours)	Cognitive Level
CO1	Identify the general characteristics and classification of plastics and to comprehend the preparation, properties and application of HDPE, LDPE & PVC	12	Understanding
CO2	Explain the preparation, properties and application of PP, PS and Acrylic plastics	11	Understanding

CO3	Paraphrase the preparation, properties and application of engineering plastics.	12	Understanding
CO4	Explain the preparation, properties and application of thermosetting plastics.	8	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Identify the general characteristics and classification of plastics and to comprehend the preparation, properties and application of HDPE, LDPE & PVC		
M1.01	Compare the advantages and disadvantages of plastics over conventional materials	2	Remembering
M1.02	Classify the plastics according to their origin and thermal response	2	Remembering
M1.03	Explain the production, properties and applications of PE	4	Understanding
M1.04	Explain production, properties and applications of PVC	4	Understanding

Contents:

Module - I

CLASSIFICATION AND COMMODITY PLASTICS

Plastics in modern life - Characteristics of plastics - History of plastics - Statistics of plastics. Comparison of plastics over other conventional materials - advantages and

disadvantages.

Classification of plastics – Based on origin, thermal response and applications.

Eco friendly plastics - examples.

Commodity plastics - Polyethylene – Monomer preparation of ethylene, polymer preparation, properties and applications of LDPE, HDPE

Properties and applications of LLDPE, XLDPPE, UHMWPE

PVC - Monomer preparation, polymer preparation, properties and applications.

Preparation of CPVC, properties and applications.

CO2	Explain the preparation, properties and application of PP, PS and Acrylic plastics		
M2.01	Explain the production, properties and applications of PP	3	Understanding
M2.02	Explain the production, properties and applications of PS and its toughened derivatives.	4	Understanding
M2.03	Explain the production, properties and applications of Acrylic plastics	4	Understanding
	Series Test - I	1	

Contents:

Module - II

Polypropylene - Isotactic, syndiotactic and atactic polypropylene-Monomer preparation, polymer preparation, properties and applications.

Polystyrene - Monomer preparation, polymer preparation, properties and applications.

Toughened polystyrene, HIPS and ABS - Expanded polystyrene.

Acrylic plastics - PMMA, PMA. Monomer preparation, polymer preparation, properties and applications.

CO3	Paraphrase the preparation, properties and application of engineering plastics.		
M3.01	Outline the monomer preparation. industrial production ,properties and application of Nylon 6and Nylon 66	3	Understanding
M3.02	Summarize the raw materials , industrial production properties ,and application of polycarbonate	3	Understanding
M3.03	Outline the raw materials industrial production properties and application of PET and PBT	3	Understanding

M3.04	Outline the raw materials industrial production ,properties and application of poly acetyl and Teflon	3	Understanding
Contents: Module - III Engineering plastics Definition - Comparison with commodity plastics - list various Engineering plastics Nylons - Different types - Nylon 6 and Nylon 6,6 - Monomer preparation, polymer preparation, properties and applications . Polyesters - PET - Monomer preparation, polymer preparation, properties and applications. PBT - Monomer preparation, polymer preparation, properties and applications. Polycarbonate - Monomer preparation, polymer preparation, properties and applications. Polyacetal and Teflon - Monomer preparation, polymer preparation, properties and application			
CO4	Explain the preparation, properties and application of thermosetting plastics		
M4.01	Compare thermosets with thermoplastics and comprehend stage A, B and C stage resins.	1	Understanding
M4.02	Explain the raw materials, industrial production, properties, moluding powder preparation and applications of PF, UF, MF	2	Understanding
M4.03	Explain the raw materials, industrial production, properties and applications of EPOXY and UPE resins.	3	Understanding
M4.04	Explain the raw materials, industrial production, properties and applications of PU.	2	Understanding
	Series Test - I	1	
Contents: Module - IV Thermo setting plastics Definition - comparison with thermoplastics - List various thermosetting plastics. Define A stage, B stage and C stage resins. Phenolic plastics – PF and RF. Novalac and Resole resins – comparison, industrial production, properties and			

applications. Molding powder preparation.

Amino plastics - UF and MF. Monomer preparation, industrial production, properties and applications.

UPE Resin - Difference between thermoplastic and thermo set polyesters , Raw materials, industrial production, properties and applications.

Common Epoxy Resin - Monomer preparation, industrial production, properties and applications.

Polyurethane - Monomer preparation, industrial production, properties and applications.

Text / Reference:

T/R	Book Title/Author
1.	Plastic materials J.A Brydson(Butterworth Heinmann, 7th Edition
2.	Plastic Product Design Engineering Hand book - Sydney Levy, J Harry DuBois (Springer US,1984)
3	Plastics Technology for Diploma Level Students and Technicians - Luqman
4	Plastics Engineering (3rd Edition) - Crawford

Online Resources:

Sl.No	Website Link
1	https://en.wikipedia.org/wiki/Engineering_plastic
2	https://www.uml.edu/Engineering/Plastics/default.aspx